Signal Processing

Understanding SIGNALS

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Hi
I'm Dr. Shailesh Sivan

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My Philosophy

Inspire Yourself, Uplift Others

Assistant Professor | Research Guide

Former Sofware Engineer| Lifetime Techie

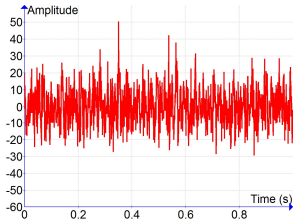
AI/ML Enthusiast | C/C++ Advocate | Algorithms & Maths

Signals

A **signal** is a **function that conveys information about the behavior or attributes of some phenomenon.**

- Oppenheim & Willsky, Signals and Systems

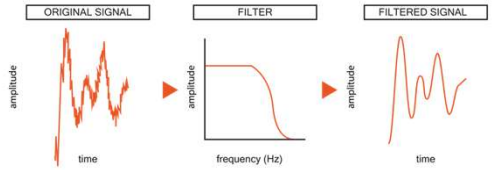
Domain	Definition of Signal	Example
Mathematics	Function over time/space	$x(t) = \sin(2\pi f t)$
DSP	Discrete-time data sequence	x[n]= [1.2, 0.9, -0.4, ...]
Electrical	Time-varying voltage/current	ECG waveform
Communications	Encoded waveform	FM signal
ML/AI	Patterned data for modeling	Speech, EEG



Signal Processing

Signal processing involves converting or transforming data in a way that allows us to see things in it that are not possible via direct observation.

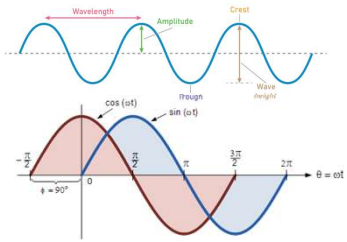
Signal processing allows engineers and scientists to **analyze, optimize, and correct signals**, including scientific **data, audio streams, images, and video.**



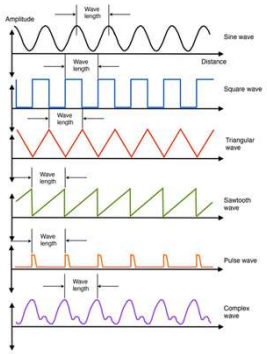
Signals as Waves

A waveform is a graphical representation of a signal as it varies with time (or another independent variable).

When a signal exhibits periodic or oscillatory behavior, it can be viewed and analyzed as a wave.



Periodic Waves



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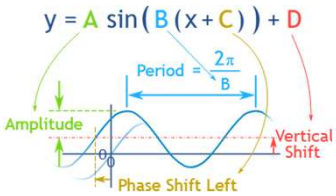
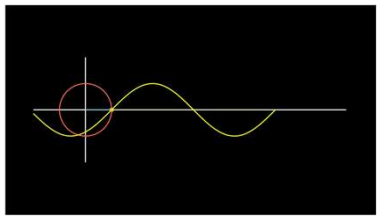
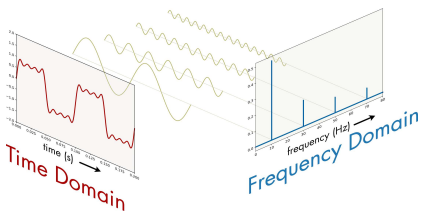
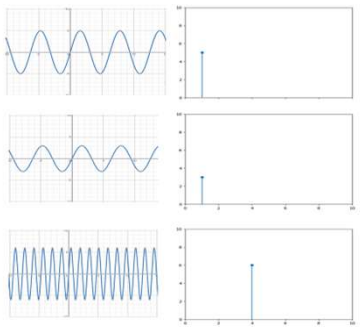




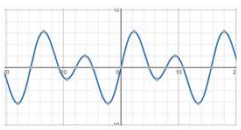
Time and Frequency Domain



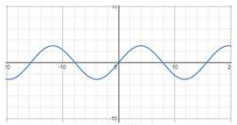


Frequency Domain Representation

How will you represent this signal in frequency domain?



y= 3 sin(0.4t) + 4 sin(0.8t)



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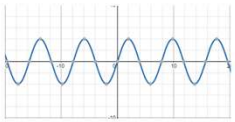
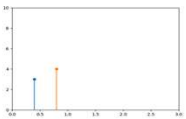


y= 4 sin(0.8t)



Frequency Domain Representation

How will you get the sinusoidal components of a signal?

Fourier Series

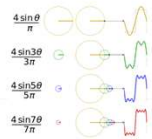
The Fourier series is used to represent a periodic function by a discrete weighted sum of sinusoidal functions

- **General Fourier Series (Real Form)**

$$f(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos\left(\frac{2\pi nt}{T}\right) + b_n \sin\left(\frac{2\pi nt}{T}\right)\right]$$

Where:

- a_0 is the average (DC) component
- a_n, b_n are Fourier coefficients
- T is the period of the signal



- **Fourier Coefficients:**

$$a_0 = \frac{1}{T}\int_0^T f(t)\,dt$$

$$a_n = \frac{2}{T}\int_0^T f(t)\cos\left(\frac{2\pi nt}{T}\right)dt \quad , \quad b_n = \frac{2}{T}\int_0^T f(t)\sin\left(\frac{2\pi nt}{T}\right)dt$$

$$f(t) = a_0 + a_1\cos\left(\frac{2\pi t}{T}\right) + b_1\sin\left(\frac{2\pi t}{T}\right) + a_2\cos\left(\frac{4\pi t}{T}\right) + b_2\sin\left(\frac{4\pi t}{T}\right) + a_3\cos\left(\frac{6\pi t}{T}\right) + b_3\sin\left(\frac{6\pi t}{T}\right) + \cdots$$

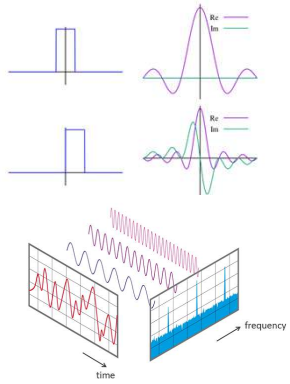
Frequency Domain Representation

Fourier Transforms

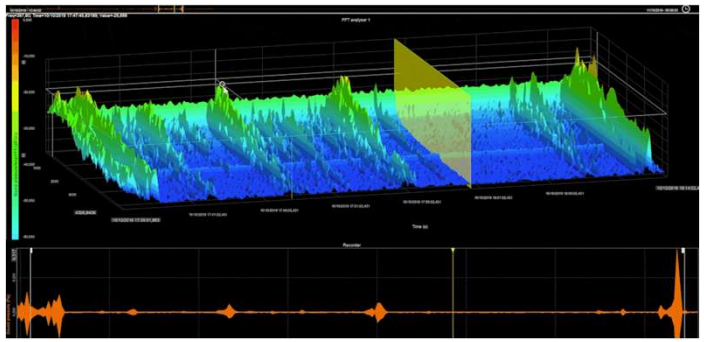
The Fourier Transform is a tool that breaks a waveform (a function or signal) into an alternate representation, characterized by sine and cosines.

The Fourier Transform shows that any waveform can be rewritten as the sum of sinusoidal functions

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\,dt$$

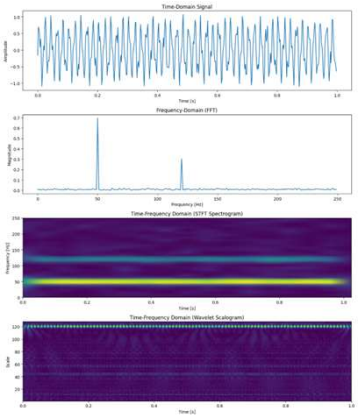


Frequency Domain Representation



Other Domains

- **Time-Domain Signal**: Shows how the signal varies with time.
- **Frequency Domain (FFT)**: Displays the dominant frequencies in the signal.
- **STFT Spectrogram**: Captures how frequency content evolves over time using short windows.
- **Wavelet Scalogram**: Offers multi-resolution time-frequency analysis, excellent for detecting transients and localized features.



Signal Processing Applications

- Audio compression and signal processing
- Data acquisition and signal processing
- Digital image and graphics processing
- Video compression and signal processing
- Speech recognition and processing
- RADAR, SONAR, and LiDAR signal processing and signal optimization

Signal Processing Applications

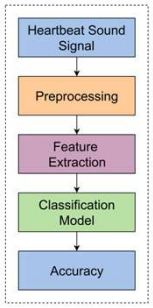
- Seismic studies and data analysis
- Geophysical applications, including oil exploration
- Data transmission, including error detection and error correction
- Economic modeling and analysis
- Medical applications, especially imagery (CAT and MRI)
- Weather forecasting
- Oceanography, including undersea acoustic performance predictions

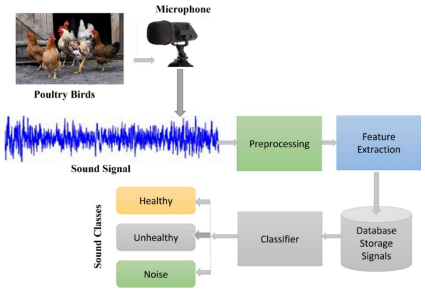
AI BASED SIGNAL PROCESSING

- Traditional signal processing: Based on transforms (Fourier, STFT, wavelets)
- AI adds pattern recognition, learning from data, and generalization
- Shift from handcrafted features to learned representations

- **Signal Denoising**
- **Signal Classification**
- **Speech Recognition**
- **Anomaly Detection in Signals**
- **Source Separation**
- **End-to-End Signal Understanding**
- **Signal Generation**

ML pipeline for SIGNAL PROCESSING





ML pipeline : FEATURE EXTRACTION

Time-Domain Features
- Mean
- Standard Deviation / Variance
- Root Mean Square (RMS)
- Zero Crossing Rate
- Skewness
- Kurtosis
- Peak Value
- Crest Factor
- Signal Energy
- Entropy

Frequency-Domain Features
- Spectral Centroid
- Spectral Bandwidth
- Spectral Roll-off
- Dominant Frequency
- Spectral Flatness
- Power Spectral Density (PSD)
- Frequency Moments

Time-Frequency Features
- Spectrogram Statistics
- Wavelet Coefficients
- Mel-Frequency Cepstral Coefficients (MFCCs)
- Chroma Features
- Delta MFCCs

Statistical & Higher-Order Features
- Autocorrelation
- Cross-correlation
- Hjorth Parameters (Activity, Mobility, Complexity)
- Fractal Dimension
- Lyapunov Exponent

Domain-Specific Features
- ECG/EEG: R-R interval, Band Power (alpha, beta, gamma)
- Speech: Pitch, Formants, Voice Activity
- Radar/Lidar: Doppler shift, Range
- Vibration: Harmonic peaks, Envelope spectrum

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